Task 3: Campus Library Search Screen

Task 3.1 – Layout managers

Use BorderLayout on the frame so the CENTER results region expands. Use FlowLayout for search controls and bottom buttons so they keep preferred sizes. Use JScrollPane around results. GridLayout would force equal cells; CardLayout is for switching views.

Task 3.2 – Nested containers

JFrame BorderLayout → NORTH JPanel BorderLayout (title NORTH, search-row JPanel FlowLayout SOUTH); CENTER JScrollPane with result list/table; SOUTH JPanel BorderLayout (button-row FlowLayout CENTER and status JLabel SOUTH).

Task 3.3 – Implementation

The source uses nested Swing panels. The central JScrollPane grows when the frame is resized, while the search and navigation rows retain preferred sizes.

Task 3.4 – Resizing tests

Test at 600×400, 900×600 and 1200×800. Confirm the result region expands and the top and bottom rows remain compact. Capture actual screenshots of the running GUI at all three sizes.

Alternative and complexity

GridBagLayout offers flexible weights and fill constraints but needs more setup. Absolute positioning with setLayout(null) and setBounds does not adapt automatically and can overlap after resize/font scaling. Nested layouts use O(c) containers for c groups and are more maintainable; both layout calculation and component storage scale with component count.

Java Source Code

```
import javax.swing.*;
import java.awt.*;
public class LibrarySearch extends JFrame {
    JTextField search=new JTextField(20);
    JComboBox<String> category=new JComboBox<>(new String[]{"All","Books","Journals","Theses"});
    DefaultListModel<String> model=new DefaultListModel<>();
    JLabel status=new JLabel("Ready");
    public LibrarySearch(){
        super("Campus Library Search");setLayout(new BorderLayout(8,8));
        JLabel title=new JLabel("Campus Library",SwingConstants.CENTER);
        JPanel north=new JPanel(new BorderLayout());north.add(title,BorderLayout.NORTH);
        JPanel row=new JPanel(new FlowLayout(FlowLayout.LEFT));
        row.add(new JLabel("Search"));row.add(search);row.add(category);
        JButton find=new JButton("Search");row.add(find);north.add(row,BorderLayout.SOUTH);
        add(north,BorderLayout.NORTH);
        model.addElement("Java Programming – Available");
        model.addElement("Database Systems – 3 copies");
        model.addElement("Operating Systems – Reference");
        add(new JScrollPane(new JList<>(model)),BorderLayout.CENTER);
        JPanel south=new JPanel(new BorderLayout());JPanel buttons=new JPanel(new FlowLayout());
        buttons.add(new JButton("Prev"));buttons.add(new JButton("Next"));buttons.add(new JButton("Reserve"));
        south.add(buttons,BorderLayout.CENTER);south.add(status,BorderLayout.SOUTH);add(south,BorderLayout.SOUTH);
```

```
        find.addActionListener(e->status.setText("Search: "+search.getText()));
        setDefaultCloseOperation(EXIT_ON_CLOSE);setMinimumSize(new Dimension(450,300));
        setSize(700,450);setLocationRelativeTo(null);
    }
    public static void main(String[] a){SwingUtilities.invokeLater(()->new LibrarySearch().setVisible(true));}
}
```

Input Files

No external input file is required; enter values in the GUI.

Output Screenshot

Illustrative console-style output in the requested black background, white text and green status theme. For GUI tasks, add genuine screenshots of the actual window and interactions.



Compile and Run

Open terminal in src, then run:
javac LibrarySearch.java
java LibrarySearch